Gaspar Zimbrek-Turkovic

Materials Scientist and Engineer · Polymers and Composites ·
Soft Materials (Hydrogels) and Sustainable Thermosets (CANs)

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Open to PhD positions and R&D roles in polymers and composites from 2027.

EDUCATION

MSc in Materials Science and Engineering
Sep 2024 – Feb 2027

École polytechnique fédérale de Lausanne (EPFL) · Lausanne, Switzerland

Minor in Management of Technology and Entrepreneurship. Expected graduation February 2027.

- **Runner-up, 2025 EPFL Innovation & Entrepreneurship in Engineering (IEE) Prize** for OptiFlow, an AI-powered wearable RFID system developed with Decathlon to improve real-time shelf inventory tracking and restocking. Responsible for hardware development and integration, prototype testing, sustainability analysis and the business case.

BSc in Materials Science and Engineering
Aug 2020 – May 2024

The Pennsylvania State University (Penn State) · University Park, PA, USA

Specialized in polymer science. GPA 3.78 / 4.0. Dean's List every semester except Fall 2020.

- **Selected to represent Penn State** at the Undergraduate Student Speaking Contest, Materials Science & Technology Technical Meeting and Exhibition (MS&T 2023), presenting research on biocompatible shape-memory triblock copolymer hydrogel fibers produced by wet spinning.
- **2nd place of 12 teams**, Materials Engineering Methodology and Design Grand Challenge, for an ammonia-based fuel cell for renewable energy storage.
- Featured in a Penn State College of Earth and Mineral Sciences video on undergraduate research.
- **Scholarships:** John and Elizabeth Holmes Teas (2022–23, 2023–24) · Penn State Metallurgy Alumni (2022–23, 2023–24) · MSE Opportunity Hancock (2023–24) · Product Evaluation Systems, Inc. (2023–24) · Robert G. and Agnes M. Faber (2022–23).

RESEARCH EXPERIENCE

Graduate Research Assistant
Oct – Dec 2026

Laboratory for Processing of Advanced Composites (LPAC), EPFL · Lausanne

Reshapable glass-fiber-reinforced dual-cure acrylate–amine vitrimer composites.

- Optimizing hot vacuum infusion parameters to improve resin impregnation and composite manufacturing.
- Investigating resin shelf life, storage stability and changes in composite thickness over time before UV-induced photopolymerization.
- Evaluating mechanical properties by tensile testing, three-point bending and interlaminar shear strength (ILSS) testing.
- Investigating chemical recycling by solvolysis in polyethylene glycol (PEG), focusing on resin degradation and recovery of the glass fibers.

Master's Thesis Research Project Feb – Aug 2026

Empa, Department of Advanced Fibers – Additives and Chemistry Group (Dr. Cédric Hervieu) · St. Gallen

Reprocessable flame-retardant thermosets based on CANs via aza-Michael addition chemistry.

- Synthesized monomers via Grignard reactions: dropwise addition, extraction, column and thin-layer chromatography, crystallization, rotary evaporation and devolatilization.
- Confirmed product structures and aza-Michael chemistry using NMR, solid-state NMR (resin), model reaction, DIP-MS and Raman spectroscopy.
- Prepared β-amino phosphine oxide epoxy (APOE) CAN resins via thermal curing.
- Evaluated flame retardancy using UL 94, cone calorimetry and PCFC testing.
- Evaluated material properties via elemental, thermal and mechanical characterization.
- Demonstrated APOE's self-healing and reprocessing capabilities using hot pressing.

EPFL Laboratory Research Project Feb – Jun 2025

Laboratory for Processing of Advanced Composites, EPFL (Prof. Véronique Michaud)

Frontal polymerization of vitrimer composites.

- Investigated thiol-acrylate vitrimer composites with transesterification-based bond exchange, synthesized via frontal polymerization, as sustainable alternatives to thermosets.
- Studied the effect of catalyst concentration (DMAP) and PET microparticle content (0–10 vol%) on thermal stability, mechanical properties and covalent bond exchange dynamics.
- Demonstrated shape memory, reprocessability and suitability for circular-economy applications.

EPFL Laboratory Research Project Sep 2024 – Jan 2025

Laboratory for Soft Materials, EPFL (Prof. Esther Amstad)

Double-network granular hydrogels for wearable resistive sensors.

- Developed double-network granular hydrogels (DNGHs) for wearable resistive sensors using PAMPS microgels and PEG-based deep eutectic solvents.
- Evaluated the impact of PEG molecular weight and hydrogen-bond acceptor type on swelling, resistivity, rheology and direct ink writing (DIW) 3D printability.
- Demonstrated shear-thinning behavior, stress recovery and successful DIW 3D printing.

Undergraduate Research Assistant Jan 2022 – May 2024

Dr. Hickey Research Group, Department of Materials Science and Engineering, Penn State

Shape-memory triblock copolymer hydrogel fibers produced via wet spinning.

- Investigated the influence of processing parameters (concentration, solvent, strain rate, needle gauge, injection speed) on hydrogel fiber processing efficiency and resulting structural and mechanical properties.
- Evaluated solvent effects (THF vs. DMF) on fiber actuator performance (actuation stroke, actuation rate and work density).
- Designed mineralization experiments using SOS hydrogel fibers to replicate collagen fibril mineralization processes.
- Analyzed crystallization behavior of strained vs. non-strained SOS hydrogel fibers.
- Developed MATLAB code to acquire and process data from water-actuation video recordings and quantify actuation kinetics of crystallized SOS fibers.

PUBLICATIONS

β-Amino phosphine oxide chemistry enables healable, reprocessable, and flame-retardant epoxy networks 2026

Hervieu, C., **Žimbrek-Turković, G.**, Cirillo, C., Rupper, P., Sekar, A., Jovic, M., Lehner, S., Gaan, S.

Preprint, SSRN · doi:10.2139/ssrn.7655948

Recyclable flame-retardant epoxy–amino ester covalent adaptable networks enabled by sustainable polyelectrolyte complexes 2026

Hervieu, C., Cirillo, C., Lehner, S., **Žimbrek-Turković, G.**, Petri, M., Rupper, P., Jović, M., Sekar, A., Grunlan, J. C., Gaan, S.

Preprint, SSRN · doi:10.2139/ssrn.7323542

TECHNICAL AND PROGRAMMING SKILLS

- **MATLAB** (data analysis and experimental processing)
- **RStudio** (linear regression modeling)
- **Origin and Proteus** (thermal data analysis)
- **Autodesk Moldflow** (injection molding simulation)
- **CrystalMaker, PyMOL and ChemDraw** (molecular visualization and chemical drawing)
- **MS Office** (Excel, Word, PowerPoint), **Adobe Photoshop**

LABORATORY EXPERIENCE – TOOLS AND SKILLS

- **Synthesis:** Organic synthesis, characterization of organic and inorganic compounds, column chromatography, kinetic experiments, crystallization, devolatilization.
- **Structural characterization:** NMR spectroscopy (Spinsolve 80 Ultra, Magritek), IR, GPC (HLC-8320GPC EcoSEC, Tosoh Bioscience), viscometry.
- **Polymerization and curing:** UV curing (Omnicure Series 2000, Analytik Jena CL-1000 UVP Crosslinker), thermal curing (Binder FD53 oven), hot pressing (Meyer/Lindenberg press).
- **Composite processing:** Hot vacuum infusion of glass-fiber-reinforced composites.
- **Sample preparation:** Centrifugation (VWR Mega Star 1.6R); grinding and polishing (Allied High Tech TwinPrep 5, MetPrep polisher, Struers Tegramin-25).
- **Optical microscopy:** Zeiss Axiovert 200, Olympus BX60M, Nikon Eclipse TS100; ImageJ and AxioVision for image analysis.
- **Flame retardancy testing:** UL 94, cone calorimetry, pyrolysis-combustion flow calorimetry (PCFC), limiting oxygen index (LOI).
- **Mechanical testing:** Of CANs, composites and hydrogel fibers: uniaxial tensile testing (MTS Criterion 43, 50 kN; universal testing machine, 125 kN) with extensometer analysis; 3-point bending (Zwick Z100, 500 N load cell); interlaminar shear strength (ILSS).
- **Thermal characterization:** TGA (PerkinElmer TGA 8000, NETZSCH TG 209 F1); DSC 214 Polyma (NETZSCH) for Tg; DMA Q800 (TA Instruments) for Tg, amplitude sweep and stress relaxation.
- **Electrical testing:** Resistance and resistivity of gels; swelling in artificial sweat (NaCl + acetic acid, pH ~5.3).

- **Rheology:** Amplitude, temperature and frequency sweeps, shear recovery and stress relaxation on Discovery HR-3 (TA Instruments) and MCR 503 (Anton Paar), TRIOS software.
- **3D printing:** Direct ink writing (CellInk Bio X bioprinter), resolution checked by optical microscopy and ImageJ; FDM printing (Original Prusa MK4 / i3 MK3, PrusaSlicer).
- **Imaging and monitoring:** FLIR T62101 infrared camera and thermocouples to follow the polymerization front of vitrimers.
- **XRD and spectroscopy:** Phase identification by XRD; elemental analysis of acoustic foam and PMN-25PT piezoelectric ceramics (Malvern Panalytical Aeris XRD, Phenom ProX SEM with EDS, ProSuite).

LANGUAGES

- Croatian (native) · English (C2) · French (B2–C1) · Italian (A2–B1)

EXTRACURRICULAR ACTIVITIES AND ACHIEVEMENTS

- **N4-ranked tennis player in Switzerland** (Interclub Nationalliga C, Montchoisi TC, 2025).
- **ATP-ranked** (doubles, 2019–20) and **ITF-ranked** (singles and doubles, 2019–21) tennis player; doubles semi-finalist (2019, 2020) and singles quarter-finalist (2020) at the Croatian National Indoor Championships.
- Winner (2016) and runner-up (2015) of the Strasbourg Write a Story Competition, non-native speaker U-18 category.